Context

Your client recently heard a presentation by the founders of Lighthouse Sanctuary, a US-based 501(c)(3) nonprofit that operates safe homes for girls who are survivors of sexual abuse or sex trafficking in the Philippines. Lighthouse Sanctuary contracts with in-country individuals and organizations to provide safehouses and rehabilitation services and is funded from local and international donors.

Inspired by the amazing good being performed by Lighthouse Sanctuary, your client has decided to create a similar organization to assist in other regions that are lacking similar services. Fortunately, Lighthouse Sanctuary has graciously shared anonymized information about their operations to help your client understand the day-to-day operations of the organization. This includes sample operational documentation such as caseload inventory records, process recording formats (structured counseling session notes), home visitation reports, and annual accomplishment report templates. Your role is to help your client develop the technology resources needed to effectively run the new organization and to help the client understand how to collect and utilize data (like that shared by Lighthouse) to drive business decisions.

The client is engaging your services so that they can meet the following primary goals before launch:

- The organization depends entirely on donations to operate, and the founders are constantly worried about **donor retention and growth**. They lose donors and don't always understand why. They run fundraising campaigns but aren't sure which ones actually move the needle versus just generating noise. They want to know which donors might give more if asked, which ones are at risk of lapsing, and how to personalize outreach without a dedicated marketing team. They also want to communicate to donors specifically how their generosity is benefiting the residents, but they don't have a good way to connect donation activity to outcomes.
- The organization's primary operational work is protecting and rehabilitating victims. The founders worry about **girls falling through the cracks**. With limited staff managing multiple safehouses, they need to know which girls are progressing and which are struggling, which interventions are actually working, and when a resident might be ready for reintegration or at risk of regression. They need a system to help them manage cases across the full lifecycle, from intake and case assessment through counseling, education, health services, and ultimately reintegration or placement. This includes structured counseling session documentation (process recordings), home visitation tracking, case conferences, and intervention planning.

- Social media is the organization's primary channel for reaching potential donors, but the founders freely admit they are **not experienced with social media**. They struggle with basic questions: What should they post? On which platforms? How often? What time of day? What kind of content actually leads to donations versus just generating likes? They have been posting sporadically and want to be more strategic, but they don't have a marketing team and can't afford to hire one. They need the technology to help them make smarter decisions about their social media presence and understand what is actually working.
- The organization needs to be able to **administer** and maintain any systems with limited staff. It needs to be easy to create, update, and (carefully) remove data from the system.
- All of this needs to be done using secure systems that **protect the privacy and safety** of victims, employees, donors, and partners. Given the extremely sensitive nature of the data — involving minors who are abuse survivors — privacy and data protection are paramount.

Your first order of business is recommending a name for this new organization.

Dataset

The data for the project is provided at the following link:

- https://drive.google.com/file/d/1Dl8AcS1ydbHKL6PU0gP6tbUPqhPsUeXZ/view?usp=sharing

The dataset includes 17 tables that are available as 17 CSV files. You are also provided a data dictionary (see Appendix A) explaining the tables and fields. These tables and fields represent what Lighthouse Sanctuary has been keeping track of and your client's new organization will need to track similar data. However, you may think of additional tables and fields to add to the database that you believe would be useful and you don't have to use all of the existing data. You can modify the data as needed to meet the needs of the project.

The dataset is organized into three main domains:

Donor and Support Domain — Tables tracking safehouses, partners, supporters, donations (monetary, in-kind, time, skills, and social media advocacy), and donation allocations. These tables support the donor-facing and administrative features of the application.

Case Management Domain — Tables tracking residents (the girls served by the organization), their case information, process recordings (counseling sessions), home visitations, education

records, health and wellbeing records, intervention plans, and incident reports. These tables support the core operational case management features of the application.

Outreach and Communication Domain — Tables tracking social media activity, engagement metrics, and public impact snapshots. These tables support the organization's social media strategy and public-facing communications.

Deliverables

IS 401 – Project Management and Systems Design

You will complete this project in four one-day sprints (ending at midnight on Monday, Tuesday, Wednesday, and Thursday), with your final deliverable and presentation due on Friday. Each day has specific deliverables for this class. You will get half of the points simply for completing each step, and the other half of points will be awarded based on the quality of your submission.

Make a copy of this Figjam board and submit the link through IS 401's Learning Suite on Monday. Fill it out throughout the week. Each day's section is due by 11:59pm that night.

Monday — Requirements (6.5 pts)

Focus on understanding the problem and planning your work.

- **Roles**: Identify a Scrum Master and Product Owner for the team.
- **Customer personas**: Create two realistic personas that represent your two most important target users. Justify why these are the most important people to solve for. You should use these throughout the week to guide your decisions.
- **Journey map**: Show the key steps users take today and identify their pain points.
- **Problem statement**: Clearly describe the specific problem your product will solve.
- **MoSCoW table**: List every requirement for INTEX in a MoSCoW table, along with at least five "nice-to-have" ideas your team comes up with. You should use this throughout the week to guide you as you prioritize your product backlog. Identify one feature you chose NOT to build and why.
- **Product backlog**: Include a clear product goal and at least 12 backlog cards.
- **Sprint "Monday" backlog**: Define a sprint goal and include at least 8 cards, each with a point estimate and exactly one person assigned. Take a screenshot of this before you start your Monday work.

- **Burndown chart**: Set up a burndown chart that will track all your progress throughout the week. You should refer to this throughout the week to see if you're on track.
- **Figma wireframe**: Brainstorm with your team how you would design the 3 most important screens for your design. Draw a wireframe for each one in Desktop view.

Tuesday — Design (4 pts)

Focus on exploring design options and deciding what to build.

- **Sprint "Tuesday" backlog**: Define a sprint goal and include at least 8 cards, each with a point estimate and exactly one person assigned. Take a screenshot of this before you start your Tuesday work.
- **AI-generated UI options**: Provide 3 screenshots of each of 3 different UI designs (9 screenshots total). Include 5 questions you asked AI about each of the 3 different UI designs and summarize in a sentence or two the takeaways from the feedback on each design.
- **Design decision**: Indicate which UI you chose. Write a short paragraph justifying why you chose it. List three changes you made to the original AI output.
- **Tech stack diagram**: Draw a diagram showing the logos for each of your chosen technologies for the frontend, backend, and database.

Wednesday — One working page (4.5 pts)

Focus on getting a real feature working end-to-end.

- **Sprint "Wednesday" backlog**: Define a sprint goal and include at least 8 cards, each with a point estimate and exactly one person assigned. Take a screenshot of this before you start your Wednesday work.
- **Current state screenshots**: Show a screenshot of at least 5 pages, in both desktop and mobile views.
- **One working page**: Demonstrate one page that is deployed to the cloud and persists data in the database.
- **User feedback**: Show your website in its current state to a real person. Choose someone who would have some insight on your target persona. Watch them use the site, listen to their feedback, and write down 5 specific things you plan to change based on what they share.
- **Burndown chart**: Share your up-to-date burndown chart to reflect your progress so far during the week.

Thursday — Iterate (5 pts)

Focus on finalizing your product and adding measurement.

- **Sprint "Thursday" backlog**: Define a sprint goal and include at least 8 cards, each with a point estimate and exactly one person assigned. Take a screenshot of this before you start your Thursday work.
- **OKR metric**: Track and display one meaningful metric in your app and explain why it is the most important measure of success for your product.
- **Accessibility**: Achieve a Lighthouse accessibility score of at least 90% on every page.
- **Responsiveness**: Ensure every page resizes appropriately for desktop and mobile views.
- **Retrospective**: Conduct a retrospective. Each person should write at least 2 things that are going well, 2 things that could have been better, and their personal greatest contribution to their team this week. Then reflect as a team on how well your solution solves the customer problem you set out to improve.

IS 413 – Enterprise Application Development

The time has come to apply all the skills that we have learned in class. While you have not done everything that is required in this project, you have the foundation to be able to dig in and figure it out using .NET and React. The web app you build will be the face of this project, so use it to impress the judges. Here are the requirements:

Build all the required pages with all the required functionality (including, but not limited to):

Public (Non-Authenticated Users)

- **Home / Landing Page:** A modern, professional landing page that introduces the organization, its mission, and provides clear calls to action for visitors to engage or support.
- **Impact / Donor-Facing Dashboard:** Displays aggregated, anonymized data showing the organization's impact (e.g., outcomes, progress, and resource use) in a clear and visually understandable way.
- **Login Page:** Allows users to authenticate using a username and password, with proper validation and error handling.

- **Privacy Policy + Cookie Consent:** Provides a privacy policy explaining data usage and includes a GDPR-compliant cookie consent notification (see the IS414 section for more information).

Admin / Staff Portal (Authenticated Users Only)

- **Admin Dashboard:** Provides a high-level overview of key metrics such as active residents across safehouses, recent donations, upcoming case conferences, and summarized progress data. Think of this as the "command center" for staff managing daily operations.
- **Donors & Contributions:** Allows staff to view, create, and manage supporter profiles, including classification by type (monetary donor, volunteer, skills contributor, etc.) and status (active/inactive). Tracks all types of contributions (monetary, in-kind, time, skills, social media) and allows staff to record and review donation activity. Supports viewing donation allocations across safehouses and program areas.
- **Caseload Inventory:** This is the core case management page. It maintains records for all residents following the structure used by Philippine social welfare agencies. Staff can view, create, and update resident profiles including demographics, case category and sub-categories (e.g., trafficked, victim of physical abuse, neglected), disability information, family socio-demographic profile (4Ps beneficiary, solo parent, indigenous group, informal settler), admission details, referral information, assigned social workers, and reintegration tracking. This page should support filtering and searching by case status, safehouse, case category, and other key fields.
- **Process Recording:** Provides forms for entering and viewing dated counseling session notes for each resident. Each process recording entry captures the session date, social worker, session type (individual or group), emotional state observed, a narrative summary of the session, interventions applied, and follow-up actions. Staff should be able to view the full history of process recordings for any resident, displayed chronologically. This is the primary tool for documenting the healing journey of each resident.
- **Home Visitation & Case Conferences:** Allows staff to log home and field visits, including the visit type (initial assessment, routine follow-up, reintegration assessment, post-placement monitoring, or emergency), observations about the home environment, family cooperation level, safety concerns, and follow-up actions. Also provides a view of case conference history and upcoming conferences for each resident.
- **Reports & Analytics:** Displays aggregated insights and trends to support decision-making. This should include donation trends over time, resident outcome metrics (education progress, health improvements), safehouse performance comparisons, and reintegration success rates. Consider structuring reports to align with

the Annual Accomplishment Report format used by Philippine social welfare agencies, which tracks services provided (caring, healing, teaching), beneficiary counts, and program outcomes.

Misc

- Any additional pages required to support functionality described in other portions of the project (e.g., security, social media, accessibility, or partner features).

Use .NET 10 / C# on the back end and React / TypeScript (Vite) on the front end. You can choose from Azure SQL Database, MySQL, or PostgreSQL for the relational database. The security database may be housed separately if you choose. Use good database principles in your design. Your app and your database must both be deployed. I would recommend using Microsoft Azure since you have the practice and the credits, but it is totally up to your team if you want to make it harder on yourself and deploy somewhere else. ;) You may modify the database as needed. Use the fields you need; don't feel obligated to display every field from every table on your UI.

Make sure to validate data and handle errors so the website is robust and reliable. Put into practice all the things we learned about writing good code. Pay attention to the details and finishing touches—titles, icons, logos, consistent look and feel, pagination, speed, and other elements that separate good websites from excellent ones.

IS 414 - Security

The IS414 portion of INTEX includes BOTH applying skills you have already learned AND learning how to implement a few new skills now that you have sufficient technical background to complete them.

In industry, features that are not documented (especially security features) don't "exist". If you would like to receive INTEX points for the security requirements, they MUST be clearly shown in your group's submitted video. Please don't spend hours implementing features and then spend seconds or minutes documenting a fraction of your work unless you would prefer to receive a fraction of the points. Appeals to grant points that are not documented at the due date will be denied.

Requirements

- Confidentiality (Encryption)
  - Use HTTPS for all public connections. It is fine if you use a subdomain and you may rely upon certificates automatically provided by the cloud provider, but you must have a valid certificate to enable TLS.
  - Redirect HTTP traffic to HTTPS.
- Authentication
  - Provide the functionality to authenticate users using a username/password (likely using ASP.NET Identity). Your identity database can be in SQLite or found in a database server.
  - Visitors (unauthenticated users) should be able to browse the home page of the site (and possibly other pages that do not need authentication)
  - Authenticated users should be able to view the page(s) described in the IS413 section.
  - Configure ASP.NET Identity (or your chosen Identity provider) to require better passwords than the default `PasswordOptions` (see https://learn.microsoft.com/en-us/aspnet/core/security/authentication/identity-configuration?view=aspnetcore-10.0 for information on the settings to change, but DO NOT follow the values suggested there). This will be STRICTLY graded according to how you were taught in class and how you were instructed to implement password policies in your lab. AI or other code/documentation suggesting policies that conflict with how you were taught in class will NOT be considered a reason to grant points.
  - All APIs should have the appropriate authentication/authorization. For example, a `/login` and `/auth/me` endpoint cannot require authentication/authorization or else they are useless, but endpoints supporting most CRUD operations (certainly all the "CUD" operations and likely some "R" operations) of the database must only be accessible/usable to authenticated and appropriately authorized users. When in doubt, make it maximally restrictive unless it breaks functionality.
- Role-Based (RBAC) Authorization
  - Only an authenticated user with an admin role should be able to add, modify, or in rare cases delete data. Only authenticated users who are donors should be able to see their donor history and the impact of those donations. Non-authenticated users without a role should be able to see some of the site (e.g., homepage, privacy policy, etc.). You may choose whether or not to have a staff (or employee) role that differs from the admin user.

- Integrity
    - Data should only be able to be changed or deleted by an authorized, authenticated user and there must be confirmation required to delete data. For example, an administrator could update or delete data because they are authorized to do so.
- Credentials
    - Handle credentials (usernames and passwords, API keys, etc.) safely. You may choose to use a secrets manager, place secrets in a separate file (e.g., an `.env` file) that is not uploaded to a code repository like GitHub or set environmental variables in your operating system. All three of these are viable options and will be granted full points. Some of these options are easier in deployment than others. You should not include credentials in your code or public repositories. Please note that having a functional site is worth more overall points than protecting credentials properly, so prioritize wisely (as much as this pains your instructor). Make it obvious how you are implementing this in your video.
- Privacy
    - Create and populate the content of a GDPR-compliant privacy policy that is linked from the footer of your site (at a minimum on the home page). You may use existing templates (see https://gdpr.eu/privacy-notice/) or generators, but the content should be tailored to your site. This will not be evaluated by a lawyer, and you should not pay for this. Current LLMs are quite adept at creating and modifying "boilerplate" privacy policies.
    - Enable a GDPR-compliant cookie consent notification. Be specific in your video about whether this is cosmetic or fully functional.
- Attack Mitigations
    - Enable the Content-Security-Policy (CSP) HTTP **HEADER**. Specify the sources you need for your site to function and no more (e.g., you might choose to define `default-src`, `style-src`, `img-src`, `script-src`, etc., but only choose what you need). It is possible to embed CSP information in a <meta> tag in the HTML, but that is not what is being evaluated here. Graders will be evaluating whether the CSP header is present in the developer tools inspector.
- Availability
    - Your site should be publicly accessible. This likely means that it is deployed to a cloud provider of your selection.
- Additional Security Features

- Some points will be allocated to adding additional security/privacy features beyond what is explicitly listed here. This allows for creativity in approach. Whatever you add, you must make clear to the TAs in your video documentation. Please briefly describe what you added and why it was added. The following list is not exhaustive but may give an idea of what types of features to consider. Some could be worth full points; most will be worth partial points depending on complexity and time required to implement and so you may choose to implement multiple additional security features.
    - Enable at least one type of third-party authentication. The third-party provider(s) are up to you.
    - Enable at least one form of two-factor or multi-factor authentication. You have flexibility in the choice and implementation. Note: You must have at least one admin and one non-admin user account without 2FA or MFA for grading purposes so we can access your site without having your phones/faces/etc.
    - Enable HTTP Secure Transport Security (HSTS) on your site. This can be tricky or trivial depending on your chosen cloud architecture. A single line of code is sufficient in some cases, but NOT in others. Don't underestimate the time required here in case your deployment is "tricky".
    - Enable a browser accessible cookie (i.e., NOT an httponly cookie) that saves a user setting that is used by React to change the page. Examples might include light/dark mode preferences, color theme preferences, language preferences, etc.
    - Enable data sanitization (for incoming data) or data encoding (for data rendered by the frontend) to help prevent injection attacks
    - Deploy both operational and identity databases to "real" DBMS (i.e., not SQLite).
    - Deploy using Docker containers instead of simply deploying to a VM
    - Or many, many other potential options

Security Rubric

Objective	Points Possible
Confidentiality - Use HTTPS/TLS	1
Confidentiality - Redirect HTTP to HTTPS	0.5

Auth - Authentication using username/password	3
Auth - Require better passwords	1
Auth - Pages and API endpoints require auth where needed	1
Auth - RBAC- Only admin user can CUD (including endpoints)	1.5
Integrity - Confirmation to delete data	1
Credentials - Stored securely and not found in public repository	1
Privacy - Privacy policy created and added to site (customized as needed)	1
Privacy - GDPR cookie consent notification fully functional	1
Attack Mitigations - CSP header set properly	2
Availability – Deployed publicly	4
Additional security features	2

IS 455 – Machine Learning

The organization you are building technology for collects rich data across multiple dimensions: donor activity, resident case histories, counseling sessions, education progress, health outcomes, home visitation reports, safehouse operations, and social media engagement. The organization's leadership wants to understand how machine learning can help them make better decisions and improve outcomes for the girls they serve.

This is your opportunity to practice **pipeline thinking**, the central idea of this course. Machine learning is not a collection of algorithms learned in isolation. It is an end-to-end decision system that begins with problem framing and ends with deployed, monitored systems. Your task is to demonstrate that you can build that complete system thoughtfully, creatively, and rigorously.

Requirements

You should deliver **as many complete machine learning pipelines as you can identify and have time to build**, integrated into or supported by the web application your team is building. Each pipeline must follow the full end-to-end ML pipeline as taught in the textbook and should address a meaningful business question for the organization. There is no maximum. The more pipelines your team delivers (with quality), the better your score. However, don't spend more time on this requirement than it's worth (1/5 of your overall INTEX grade).

Each pipeline should demonstrate the full lifecycle as outlined in the Foreword:

1. **Problem Framing** (Ch. 1): Define the business question, select success metrics, and determine whether a predictive or explanatory approach is appropriate. This is one of the most important decisions you will make. As the textbook explains, *prediction* and *explanation* are two distinct modeling goals that are related but not interchangeable. Predictive modeling aims to generate reliable predictions judged by performance on new, unseen data. Explanatory modeling aims to understand and quantify relationships between variables, ideally identifying cause-and-effect patterns that can inform strategy or policy. Confusing these two goals is a common source of costly analytical mistakes. For each pipeline, generate both a causal and predictive model. Besides deploying a predictive pipeline, indicate within the .ipynb file which features are most impactful and recommend decisions that can be made based on the results

2. **Data Acquisition and Preparation** (Ch. 2–5, 7): Identify and collect the data needed from the provided tables. Clean, transform, and engineer features from raw data. As the textbook notes, data preparation is often the most time-intensive stage of any

analytics project. Expect to spend 60–80% of your pipeline effort here. Build reproducible data preparation pipelines (Ch. 7), not one-off scripts.

3. **Exploration** (Ch. 6, 8): Examine distributions, relationships, and anomalies to build intuition about the data before modeling. This is where you discover what the data can actually tell you. Look for the relationships that have been built into this dataset. They are there to be found.

4. **Modeling** (Ch. 9–14): Select and train algorithms that learn patterns from the data. You may use any approach covered in the course: multiple linear regression for causal or predictive purposes (Ch. 9–11), decision trees (Ch. 12), classification (Ch. 13), or ensemble methods like bagging, boosting, and stacking (Ch. 14). Remember that for explanatory work, a model that sacrifices some predictive accuracy to produce clear, defensible estimates, like a carefully specified OLS regression, can be preferable to a black-box model with marginally better predictions. For predictive work, complex features and less interpretable models can be appropriate when they improve accuracy and operational performance.

5. **Evaluation and Selection** (Ch. 15): Assess model performance using appropriate metrics, validation strategies, and fairness checks. Use proper train/test splits or cross-validation. Tune hyperparameters. Compare multiple models where appropriate. Most importantly, interpret your results *in business terms*. Don't just report an R^2 or accuracy score. Explain what the numbers mean for the organization's decisions.

6. **Feature Selection** (Ch. 16): Thoughtfully select which features to include. Demonstrate that you understand why certain features matter and others don't. Use feature importance, selection techniques, or domain reasoning to justify your choices.

7. **Deployment** (Ch. 17): Move your validated model into the production web application where it can generate predictions, automate decisions, or inform stakeholders. This could mean serving predictions through an API endpoint, displaying model outputs on a dashboard, or providing an interactive interface where users can input data and receive predictions or insights.

Each pipeline should address a **different business problem**. Teams should try to demonstrate a variety of modeling approaches across their pipelines; ideally including at least one predictive model and one explanatory model, but this is not strictly required. The dataset has been designed with multiple discoverable relationships across the donor domain, the case

management domain, and the social media/outreach domain. Look carefully at the data and think about what questions the organization would want answered.

Why These Instructions Are Intentionally Vague

In the real world, nobody hands you a target variable and tells you which algorithm to use. A business leader says “we need to understand our donors better” or “we’re worried about girls falling through the cracks” and it is *your job* as the data scientist to translate that into a concrete, solvable problem. As the textbook describes in the Career Pathways section, effective practitioners at the problem framing stage “determine which questions are worth answering, select success metrics, and translate context into requirements technical teams can implement.” The ability to look at a dataset and see opportunities, to ask “what questions could this data answer?”, is one of the most valuable skills you will develop. This project is designed to simulate that experience.

You are not being told what to predict, what to classify, or what to explain. The data has been carefully constructed with real, discoverable relationships across multiple domains. Your task is to find them. You will be evaluated relative to each other, and the teams that identify more meaningful opportunities and execute them well will earn higher scores.

How to Use AI Effectively

You are encouraged to use AI tools throughout this process, but *how* you use AI matters. We recommend using AI in the following sequence, which mirrors the pipeline stages you have learned:

1. **AI as Problem Setter** (supports Problem Framing). Start by using AI to help you understand the business domain and brainstorm potential problems worth solving. Give AI the data dictionary and ask it to help you think about what questions the organization might need answered. Push it to consider both predictive questions (“Can we forecast which residents are ready for reintegration?”) and explanatory questions (“What factors most drive successful reintegration?”). Don’t accept its first suggestions uncritically. Push it to think deeper about the business context and to consider perspectives you might have missed.

2. **AI as Creative Expander** (supports Data Preparation and Exploration). Once you have a candidate problem, use AI to help you think about features you might engineer, relationships you might explore, and data from other tables you might join. Ask it to suggest transformations, interaction terms, or aggregations you might not have

considered. Remember that 60–80% of pipeline effort is data preparation. This is where AI can help you think more broadly about the raw materials available to you.

3. **AI as Critical Evaluator** (supports Evaluation and the prediction/explanation tradeoff). Before you commit to a direction, use AI to poke holes in your approach. Ask it: "What could go wrong with this model? Am I confusing prediction with explanation?" Use it to stress-test your thinking the way a skeptical colleague would. For explanatory models, ask whether your causal claims are defensible. For predictive models, ask about data leakage and validation soundness.

4. **AI as Verification Agent** (supports Evaluation and Selection). Use AI to verify that your code is correct, your evaluation methodology is sound, and your results make sense. Ask it to review your train/test split, validate your metrics, and confirm that your feature selection and hyperparameter tuning are reasonable.

5. **AI as Production Agent** (supports Deployment). Finally, use AI to help you deploy your model, create API endpoints, build dashboard visualizations, and integrate your results into the web application. This is where AI can save you the most time on implementation details so you can focus on ensuring your deployed model provides real value to end users.

The key principle: *you* drive the thinking and decision-making at every stage. AI accelerates your work, but the intellectual ownership must be yours. You must understand and be able to explain every part of your pipeline. If you cannot explain why you made a particular choice, you are using AI incorrectly. The goal, as the textbook states, is "not just to fit models, but to build analytical capability that holds up in real environments."

Applying the Principles from the Textbook

Your pipelines will be evaluated on how well they apply the principles taught in Chapters 1–17. Specifically, we will be looking for evidence that you understand:

- **Pipeline thinking over algorithm-only thinking** (Foreword, Ch. 1): Each pipeline should tell a complete story from business problem to deployed solution, not just demonstrate that you can run an algorithm.

- **The prediction vs. explanation distinction** (Foreword, Ch. 9–11): For each pipeline, you must explicitly state whether your goal is prediction or explanation and make modeling choices that are consistent with that goal. If you are explaining, coefficients and interpretability matter more than predictive accuracy. If you are predicting,

out-of-sample performance matters more than whether individual coefficients are interpretable. Do not treat accurate predictions as causal evidence, and do not produce interpretable analyses that cannot be operationalized.

- **Rigorous data preparation** (Ch. 2–3, 7): Feature engineering, handling missing data, building reproducible pipelines. The quality of what goes into the model matters as much as the model itself.
- **Thoughtful exploration** (Ch. 6, 8): Demonstrate that you actually looked at the data before modeling. Distributions, correlations, anomalies, and relationships should be documented and should inform your modeling choices.
- **Appropriate model selection** (Ch. 12–14): Use the right tool for the job. Consider multiple approaches. If you use an ensemble (Ch. 14), explain why it is appropriate. If you use a decision tree (Ch. 12), explain what you gain in interpretability.
- **Evaluation discipline** (Ch. 15): Proper validation, appropriate metrics, honest interpretation of results. Understand the real-world consequences of errors for this specific organization.
- **Feature selection with purpose** (Ch. 16): Don't throw everything in and hope for the best. Demonstrate that you thought about which features matter and why.
- **Deployment as a first-class concern** (Ch. 17): A model that only exists in a notebook is not a pipeline. Move it into production.

A team that demonstrates mastery of these principles across multiple pipelines will outscore a team that produces many pipelines with shallow execution. Quality and quantity both matter, but quality comes first.

Rubric (20 points total)

Your score will be based on the number of complete, quality pipelines your team delivers. Each pipeline will be evaluated using the following criteria aligned with the full ML pipeline:

Pipeline Stage	What We're Evaluating
Problem Framing	Is the business problem clearly stated? Does it matter to the organization? Is the choice of predictive vs. explanatory approach explicitly justified?

Data Acquisition, Preparation & Exploration	Is the data explored thoroughly? Are missing values, outliers, and feature engineering handled well? Is the data preparation reproducible as a pipeline? Are joins across tables done correctly and documented?
Modeling & Feature Selection	Is the model appropriate for the stated goal (prediction or explanation)? Are multiple approaches considered or compared? Is feature selection thoughtful and justified?
Evaluation & Selection	Are appropriate metrics used? Is the model validated properly (e.g., train/test split, cross-validation)? Are results interpreted in business terms, not just statistical terms?
Deployment & Integration	Is the model deployed and accessible? Is it integrated with the web application in a meaningful way? Does it provide value to end users?

What to Submit

For each pipeline, submit a Jupyter notebook (.ipynb) in your team's GitHub repository inside a folder named ml-pipelines/. Name each notebook descriptively (e.g., donor-churn-classifier.ipynb, reintegration-readiness.ipynb). Each notebook should be self-contained and tell the complete story of that pipeline from problem framing through deployment. Specifically, each notebook must include the following sections:

1. **Problem Framing.** A clear written explanation (not just code) of the business problem you are solving, who in the organization cares about it, and why it matters. Explicitly state whether your approach is predictive or explanatory and justify that choice using the framework from the textbook.

2. **Data Acquisition, Preparation & Exploration.** Load the relevant data, explore it visually and statistically, document what you find (distributions, correlations, missing values, outliers), and prepare it for modeling. Show your feature engineering decisions and explain why you made them. If you join data from multiple tables, explain the logic. Build this as a reproducible pipeline (Ch. 7), not a one-off script.

3. **Modeling & Feature Selection.** Build your model(s), compare approaches where appropriate, and document your choices. Justify your feature selection. If you try multiple algorithms, show the comparison. Include hyperparameter tuning if relevant. If your goal is explanatory, discuss what the model structure reveals about relationships in the data. If your goal is predictive, focus on out-of-sample performance.

4. **Evaluation & Interpretation.** Evaluate your model using appropriate metrics. Use proper validation (train/test split, cross-validation). Interpret your results *in business terms*, not just reporting an R^2 or accuracy score; explain what it means for the organization's decisions. Discuss the real-world consequences of false positives and false negatives for this specific context.

5. **Causal and Relationship Analysis.** This is critical. For each pipeline, include a written analysis discussing the relationships you discovered in the data. What features are most important and why? Do the relationships make theoretical sense? If your model is explanatory, what causal story do the coefficients or feature importances tell? Are those causal claims defensible, or are you observing correlation? If your model is predictive, what does it reveal about the underlying data structure even if the goal isn't causal inference? Be honest about the limitations: correlation is not causation, and you should discuss when you can and cannot make causal claims. This section is where you demonstrate that you understand the prediction vs. explanation distinction at a deep level.

6. **Deployment Notes.** Briefly describe how this model is deployed and integrated into the web application (e.g., API endpoint, dashboard component, interactive form). Include any relevant code snippets or references to where the integration code lives in the repo.

The notebook should be fully executable. A TA should be able to run it top to bottom and reproduce your results. Make sure your data paths are correct relative to the repository structure.

In addition to the notebooks, your deployed web application should include the model outputs in a meaningful way (predictions, dashboards, interactive tools, etc.) as described in the Deployment & Integration section of the rubric.

Important Notes

- Each pipeline must address a genuinely different business problem, not just the same problem with different algorithms.
- A poorly executed pipeline will not help your score. Focus on doing each pipeline well before starting the next one.
- Notebooks that are not executable or that lack written analysis will receive significant deductions regardless of model quality.

Final Deliverable Submission

Deliverable Due Date: Friday April 10 at 10:00 am (the morning of the presentations).
Deliverable Submission Location: https://byu.az1.qualtrics.com/jfe/form/SV_bsjPxSQyEdIQRhA
Deliverable Content:

1. **Group Information -** Please provide the correct group number and group members.
2. **URLs -** Include the correct URLs to your Website, GitHub repository (linked to the correct branch), ipynb files for each pipeline (or a screenshot of your Azure ML Designer view), and video walkthroughs. Grading will go VERY badly if you make typos here.
    1. Please ensure when submitting that your GitHub repository is set to Public for grading purposes.
    2. **Video Summaries -** Each team should submit a link to the videos (one for each class) that go through the requirements of IS413, IS414, and IS455. Please be succinct in your coverage of each feature. Graders get grumpy and stop granting points when submitted videos ramble. Submit a link to the video hosted PUBLICLY (e.g., Box, Dropbox, YouTube, etc.). Your video must be accessible to TAs and Instructors.
        1. This video should provide ALL information that the graders need to evaluate your solution. The video should demonstrate that you have completed the requirements outlined in this document. If you have not completed something, be forthright about it in the video. Do not try to pass off unfinished requirements as complete. This is considered academic misconduct. You do not need to have all team members present or even show yourself in the video. Be thorough, but concise.
        2. Videos do not need to be professional, but they should clearly demonstrate the completion of requirements (at high enough resolution) and succinctly. If you record the wrong screen in your video and the graders cannot see the features, no points will be granted. Please check your videos before submitting.
        3. The videos must be available to all who have the link (i.e., public or unlisted, but not private)
        4. Each video should address the requirements for that class only. For example, the IS455 TAs will not be viewing the videos for IS413 and vice versa.
        5. If you do not demonstrate the completion of a requirement in your video, it will be considered missing. “We finished it but forgot to put it in the video …” or “...but we showed that feature in the video for another class…” will not be considered acceptable reasons to grant points for items not demonstrated in the videos.
3. **User Credentials** - While you need to be very careful with credentials, please provide faculty and TAs the login credentials to your Web application. You will need to provide the usernames/passwords for the following:

1. An admin user that does NOT have multi-factor authentication (MFA) enabled.
2. A donor user that does NOT have multi-factor authentication (MFA) enabled but is connected with historical donations.
3. An account (admin or donor) that DOES have multi-factor authentication (MFA) enabled. We won't be able to get into this account, but we will use it to test whether MFA is required or not.

The fastest way to upset your graders is making them do extra work to find the correct URLs or log in to your site. Points will be lost if we have to reach out for more information. Maximize your score by keeping your graders happy.

Peer Evaluation

You MUST complete the INTEX2 Peer Eval by **Friday April 10th at 11:59 PM**. The survey can be found here:

https://byu.az1.qualtrics.com/jfe/form/SV_7VXtQGm7rT4cvoa

Please complete this immediately after your presentation. You will not receive an INTEX grade if you do not complete your peer evaluation. The results of your eval will be kept confidential, but it will affect team member grades. We want this to be fair for those that did the work. The solution is to make sure that you pull your weight and more. Help the team and be a great team member helping where needed and doing quality work!

Presentation

You will present your solution to a panel composed of industry professionals, faculty, and possibly TAs. Your solution will be evaluated on the quality of communication of the presentation, the effectiveness of your technology demo, your creativity in approaching the business problem, and how satisfied the clients are with your overall solution. The presentation will be worth 20% of the overall INTEX score.

You will be randomly assigned a presentation slot. Presentations will be held Friday beginning at 12:00pm. A schedule with the assigned times for each group will be posted. Each presentation period will be structured as follows:

Presentation and Tech Demo	20 minutes (Student group presents)
Questions & Answers	5 minutes (Student group answers questions)
Judge Deliberation	5 minutes (Student group waits in the hall)

Feedback	5-10 minutes (Student group comes back to room)
Break and Next Team Setup	5 minutes

The judges for your presentation will take the role of the client. Your role is to pitch your system/solution as worthy of continued investment and/or adoption by the client. Get them excited about the system and how it meets their needs. The judging panel will be evaluating how your system works, how and whether it meets the client's needs, the quality of the presentation, and whether you creatively went beyond the minimum requirements. The judges have technical expertise but will primarily evaluate the business case for investing in your solution.

This is a business presentation with a substantive tech demo. The tech demo is likely the most important part of the presentation. Prioritize showing us your solution. The demo shouldn't be the only part of your presentation, but it is the most important part. It is probably useful to highlight the business problems and how you have solved them, but please don't spend the majority of the presentation telling the client the case background.

Do NOT bring handouts to the presentation.

Logistics

Presentation Schedule

Please double check your room/time on presentation day. It is your responsibility to arrange a swap with another team if you are unhappy with your presentation time. Include BOTH teams in a Slack DM to Professor Wells to initiate a conversation about a swap.

Slack

A Slack workspace has been set up for you to communicate with instructors, TAs, the clients, and other students (see below). Please use this link:

https://join.slack.com/t/intexw26/shared_invite/zt-3udumsa9x-P87AgyhpD_DDG64adcimmg

You will be able to create your own channels for your team within this workspace.

Questions

You are not able to ask Faculty and TAs for help with technical problems, but you are welcome to ask questions in regard to the case itself or the requirements. However, due to the number of groups and students involved, please post any questions on the ***#questions*** channel in our INTEX Slack workspace so that everybody can benefit from the response and that the faculty are not answering the same question over and over again to different groups.

You may ask questions to other students working on the project (use the “questions” channel in our slack workspace), but you may not ask questions of former students who have completed a similar project in a past INTEX. Seeking help from previous students in any way or using any of their material would be considered cheating. You may ask questions of people in industry or other outside help, but the work must be your own.

Appendix A: Data Dictionary

Table Overview

Table	Detail	Purpose
`safehouses`	One row per safehouse	Physical locations where girls are housed and served
`partners`	One row per partner	Organizations and individuals who deliver services
`partner_assignments`	One row per partner × safehouse × program area	Which partners serve which safehouses and in what capacity
`supporters`	One row per donor/supporter	People and organizations who donate money, goods, time, skills, or advocacy
`donations`	One row per donation event	Individual donations across all types
`in_kind_donation_items`	One row per line item	Item-level details for in-kind donations
`donation_allocations`	One row per donation × safehouse × program area	How donation value is distributed across safehouses and program areas
`residents`	One row per resident	Case records for girls currently or formerly served
`process_recordings`	One row per counseling session	Dated counseling session notes for each resident
`home_visitations`	One row per home/field visit	Home and field visit records for residents and families
`education_records`	One row per resident per month	Monthly education progress and attendance
`health_wellbeing_records`	One row per resident per month	Monthly physical health, nutrition, sleep, and energy
`intervention_plans`	One row per intervention/goal	Individual intervention plans, goals, and services
`incident_reports`	One row per incident	Individual safety and behavioral incident records

social_media_posts	One row per social media post	Organization social media activity, content, and engagement metrics (API-like schema)
safehouse_monthly_metrics	One row per safehouse per month	Aggregated monthly outcome metrics for each safehouse
public_impact_snapshots	One row per month	Anonymized aggregate impact reports for public/donor communication

safehouses

Safehouse locations operated by the organization.

Field	Type	Description
safehouse_id	integer	Primary key
safehouse_code	string	Human-readable short code (e.g., SH-01)
name	string	Display name
region	string	One of: Luzon, Visayas, Mindanao
city	string	City of the safehouse
province	string	Province of the safehouse
country	string	Always Philippines
open_date	date	Date the safehouse opened
status	string	Active or Inactive
capacity_girls	integer	Maximum number of girls the facility can house
capacity_staff	integer	Target on-site staff headcount
current_occupancy	integer	Current number of girls housed
notes	string	Free-text notes

partners

Organizations and individuals contracted to deliver services (education, operations, transport, etc.).

Field	Type	Description
partner_id	integer	Primary key
partner_name	string	Full name or organization name
partner_type	string	Organization or Individual
role_type	string	One of: Education, Evaluation, SafehouseOps, FindSafehouse, Logistics, Transport, Maintenance
contact_name	string	Primary contact person
email	string	Contact email address
phone	string	Contact phone number
region	string	Primary region served

status	string	Active or Inactive
start_date	date	Contract start date
end_date	date	Contract end date; null if still active
notes	string	Free-text notes

partner_assignments

Assignments of partners to safehouses and program areas.

Field	Type	Description
assignment_id	integer	Primary key
partner_id	integer	FK → partners.partner_id
safehouse_id	integer	FK → safehouses.safehouse_id (nullable)
program_area	string	One of: Education, Wellbeing, Operations, Transport, Maintenance
assignment_start	date	Assignment start date
assignment_end	date	Assignment end date; null if active
responsibility_notes	string	Description of the assignment
is_primary	boolean	Whether this is the partner's primary assignment
status	string	Active or Ended

supporters

Donors, volunteers, skilled contributors, and partner organizations that provide support.

Field	Type	Description
supporter_id	integer	Primary key
supporter_type	string	One of: MonetaryDonor, InKindDonor, Volunteer, SkillsContributor, SocialMediaAdvocate, PartnerOrganization
display_name	string	Name shown in communications
organization_name	string	Organization name (null for individuals)
first_name	string	First name (null for organizations)
last_name	string	Last name (null for organizations)
relationship_type	string	One of: Local, International, PartnerOrganization
region	string	Region of record
country	string	Country of the supporter
email	string	Contact email address
phone	string	Contact phone number
status	string	Active or Inactive
first_donation_date	date	Date of the supporter's first donation (nullable)

acquisition_channel	string	How the supporter first learned about the organization. One of: Website, SocialMedia, Event, WordOfMouth, PartnerReferral, Church
created_at	datetime	When the supporter record was created

donations

Donation events across all donation types.

Field	Type	Description
donation_id	integer	Primary key
supporter_id	integer	FK → supporters.supporter_id
donation_type	string	One of: Monetary, InKind, Time, Skills, SocialMedia
donation_date	date	Date of the donation
channel_source	string	One of: Campaign, Event, Direct, SocialMedia, PartnerReferral
currency_code	string	PHP for monetary (in Philippine pesos); null otherwise
amount	decimal	Monetary amount; null for non-monetary
estimated_value	decimal	Value in the unit given by impact_unit
impact_unit	string	One of: pesos, items, hours, campaigns
is_recurring	boolean	Whether this donation is part of a recurring commitment
campaign_name	string	Name of the fundraising campaign, if applicable (nullable). Examples: Year-End Hope, Back to School, Summer of Safety, GivingTuesday
notes	string	Free-text notes
created_by_partner_id	integer	FK → partners.partner_id (nullable)
referral_post_id	integer	FK → social_media_posts.post_id (nullable). Populated for donations where channel_source is SocialMedia, linking to the post that likely referred the donation

in_kind_donation_items

Line-item details for in-kind donations.

Field	Type	Description
item_id	integer	Primary key
donation_id	integer	FK → donations.donation_id
item_name	string	Item description

item_category	string	One of: Food, Supplies, Clothing, SchoolMaterials, Hygiene, Furniture, Medical
quantity	integer	Quantity donated
unit_of_measure	string	One of: pcs, boxes, kg, sets, packs
estimated_unit_value	decimal	Estimated value per unit in Philippine pesos (PHP)
intended_use	string	One of: Meals, Education, Shelter, Hygiene, Health
received_condition	string	One of: New, Good, Fair

donation_allocations

How donations are allocated across safehouses and program areas.

Field	Type	Description
allocation_id	integer	Primary key
donation_id	integer	FK → donations.donation_id
safehouse_id	integer	FK → safehouses.safehouse_id
program_area	string	One of: Education, Wellbeing, Operations, Transport, Maintenance, Outreach
amount_allocated	decimal	Allocated value in Philippine pesos (PHP)
allocation_date	date	Date of allocation
allocation_notes	string	Free-text notes

residents

Case records for girls currently or formerly served by the organization. This table is modeled after real caseload inventory documentation used by Philippine social welfare agencies.

Field	Type	Description
resident_id	integer	Primary key
case_control_no	string	Case control number (e.g., C0073)
internal_code	string	Anonymized internal identifier
safehouse_id	integer	FK → safehouses.safehouse_id
case_status	string	One of: Active, Closed, Transferred
sex	string	F (all residents in this program are female)
date_of_birth	date	Date of birth
birth_status	string	One of: Marital, Non-Marital
place_of_birth	string	City/municipality of birth
religion	string	Religious affiliation
case_category	string	One of: Abandoned, Foundling, Surrendered, Neglected
sub_cat_orphaned	boolean	Is the child orphaned?

sub_cat_trafficked	boolean	Is the child a trafficked child?
sub_cat_child_labor	boolean	Is the child a victim of child labor?
sub_cat_physical_abuse	boolean	Is the child a victim of physical abuse?
sub_cat_sexual_abuse	boolean	Is the child a victim of sexual abuse?
sub_cat_osaec	boolean	Is the child a victim of OSAEC/CSAEM?
sub_cat_cicl	boolean	Is the child in conflict with the law (CICL)?
sub_cat_at_risk	boolean	Is the child at risk (CAR)?
sub_cat_street_child	boolean	Is the child in a street situation?
sub_cat_child_with_hiv	boolean	Is the child living with HIV?
is_pwd	boolean	Is the child a person with a disability?
pwd_type	string	Type of disability if applicable (nullable)
has_special_needs	boolean	Does the child have mental/health/developmental needs?
special_needs_diagnosis	string	Diagnosis/type if applicable (nullable)
family_is_4ps	boolean	Is the family a 4Ps (Pantawid Pamilyang Pilipino Program) beneficiary?
family_solo_parent	boolean	Is the parent a solo parent?
family_indigenous	boolean	Is the family part of an indigenous group?
family_parent_pwd	boolean	Is the parent a person with a disability?
family_informal_settler	boolean	Is the family an informal settler or homeless?
date_of_admission	date	Date admitted to the safehouse
age_upon_admission	string	Age at admission (e.g., 13 Years 2 months), current data may not be calculated accurately.
present_age	string	Current age as of reporting date, current data may not be calculated accurately
length_of_stay	string	Duration in the center (e.g., 3 Years 1 months), current data may not be calculated accurately
referral_source	string	One of: Government Agency, NGO, Police, Self-Referral, Community, Court Order
referring_agency_person	string	Name of referring agency or person
date_colb_registered	date	Date Certificate of Live Birth (COLB) was registered (nullable)
date_colb_obtained	date	Date Certificate of Live Birth (COLB) was obtained from Local Civil Registry (LCR) or Philippine Statistics Authority (PSA) (nullable)
assigned_social_worker	string	Name(s) of assigned social worker(s)
initial_case_assessment	string	Initial assessment/plan (e.g., For Reunification, For Foster Care)
date_case_study_prepared	date	Date the child case study report was prepared (nullable)

reintegration_type	string	One of: Family Reunification, Foster Care, Adoption (Domestic), Adoption (Inter-Country), Independent Living, None (nullable)
reintegration_status	string	One of: Not Started, In Progress, Completed, On Hold (nullable)
initial_risk_level	string	Risk level assessed at intake. One of: Low, Medium, High, Critical
current_risk_level	string	Most recently assessed risk level. One of: Low, Medium, High, Critical
date_enrolled	date	Same as date_of_admission (for compatibility)
date_closed	date	Date the case was closed; null if still open
created_at	datetime	Record creation timestamp
notes_restricted	string	Restricted-access free-text field

process_recordings

Structured counseling session notes for each resident, following the “Process Recording” format used by Philippine social welfare practitioners. Each entry documents a dated interaction between a social worker and a resident, including observations, interventions, and follow-up actions.

Field	Type	Description
recording_id	integer	Primary key
resident_id	integer	FK → residents.resident_id
session_date	date	Date of the counseling session
social_worker	string	Name of the social worker conducting the session
session_type	string	One of: Individual, Group
session_duration_minutes	integer	Duration of the session in minutes
emotional_state_observed	string	Emotional state at the start of the session. One of: Calm, Anxious, Sad, Angry, Hopeful, Withdrawn, Happy, Distressed
emotional_state_end	string	Emotional state at the end of the session. Same enum as emotional_state_observed
session_narrative	string	Narrative summary of the session (what was discussed, what was observed)
interventions_applied	string	Description of interventions or techniques used
follow_up_actions	string	Planned follow-up actions
progress_noted	boolean	Whether progress was noted in this session
concerns_flagged	boolean	Whether any concerns were flagged
referral_made	boolean	Whether a referral was made to another professional (e.g., psychologist, legal)
notes_restricted	string	Restricted-access free-text field

home_visitations

Records of home and field visits conducted by social workers to the families or guardians of residents. Home visitations are a critical part of case assessment, reintegration planning, and post-placement monitoring.

Field	Type	Description
`visitation_id`	integer	Primary key
`resident_id`	integer	FK → residents.resident_id
`visit_date`	date	Date of the visit
`social_worker`	string	Name of the social worker who conducted the visit
`visit_type`	string	One of: Initial Assessment, Routine Follow-Up, Reintegration Assessment, Post-Placement Monitoring, Emergency
`location_visited`	string	Location or address visited
`family_members_present`	string	Description of who was present (e.g., “Mother and aunt”)
`purpose`	string	Purpose of the visit
`observations`	string	Narrative observations about the home environment and family
`family_cooperation_level`	string	One of: Highly Cooperative, Cooperative, Neutral, Uncooperative
`safety_concerns_noted`	boolean	Whether safety concerns were noted during the visit
`follow_up_needed`	boolean	Whether follow-up action is needed
`follow_up_notes`	string	Details of required follow-up (nullable)
`visit_outcome`	string	One of: Favorable, Needs Improvement, Unfavorable, Inconclusive

education_records

Monthly education progress records for each resident, tracking enrollment in educational programs, attendance, and academic progress.

Field	Type	Description
`education_record_id`	integer	Primary key
`resident_id`	integer	FK → residents.resident_id
`record_date`	date	Date of the record
`program_name`	string	One of: Bridge Program, Secondary Support, Vocational Skills, Literacy Boost

course_name	string	One of: Math, English, Science, Life Skills, Computer Basics, Livelihood
education_level	string	One of: Primary, Secondary, Vocational, CollegePrep
attendance_status	string	One of: Present, Late, Absent
attendance_rate	decimal	Rolling attendance rate (0.0–1.0)
progress_percent	decimal	Overall program progress (0–100)
completion_status	string	One of: NotStarted, InProgress, Completed
gpa_like_score	decimal	Grade-like composite (1.0–5.0 scale)
notes	string	Free-text notes

health_wellbeing_records

Monthly physical health and wellbeing records for each resident, including medical, dental, and nutritional assessments.

Field	Type	Description
health_record_id	integer	Primary key
resident_id	integer	FK → residents.resident_id
record_date	date	Date of the record
weight_kg	decimal	Body weight in kilograms
height_cm	decimal	Height in centimeters
bmi	decimal	Body mass index
nutrition_score	decimal	Nutrition quality score (1.0–5.0)
sleep_score	decimal	Sleep quality score (1.0–5.0)
energy_score	decimal	Daytime energy score (1.0–5.0)
general_health_score	decimal	Overall health score (1.0–5.0)
medical_checkup_done	boolean	Whether a medical check-up was conducted this period
dental_checkup_done	boolean	Whether a dental check-up was conducted this period
psychological_checkup_done	boolean	Whether a psychological check-up was conducted this period
medical_notes_restricted	string	Restricted-access free-text field

intervention_plans

Individual intervention plans, goals, and services provided to each resident. Each plan represents a structured goal with a target area, description, and progress tracking. Plans are created during case conferences and updated during process recording sessions.

Field	Type	Description
plan_id	integer	Primary key

resident_id	integer	FK → residents.resident_id
plan_category	string	One of: Safety, Psychosocial, Education, Physical Health, Legal, Reintegration
plan_description	string	Description of the intervention or goal
services_provided	string	Services provided (e.g., Caring, Healing, Teaching, Legal Services)
target_value	decimal	Numeric target for the goal on the relevant scale (nullable)
target_date	date	Target date for achievement
status	string	One of: Open, In Progress, Achieved, On Hold, Closed
case_conference_date	date	Date of the case conference where this plan was created/reviewed (nullable)
created_at	datetime	Record creation timestamp
updated_at	datetime	Last update timestamp

incident_reports

Individual safety and behavioral incident records for residents. Each row represents a specific incident that was observed, reported, or documented by staff. Incident data provides granular detail beyond the monthly aggregate counts in safehouse_monthly_metrics.

Field	Type	Description
incident_id	integer	Primary key
resident_id	integer	FK → residents.resident_id
safehouse_id	integer	FK → safehouses.safehouse_id
incident_date	date	Date of the incident
incident_type	string	One of: Behavioral, Medical, Security, RunawayAttempt, SelfHarm, ConflictWithPeer, PropertyDamage
severity	string	One of: Low, Medium, High
description	string	Narrative description of the incident
response_taken	string	Description of the staff response
resolved	boolean	Whether the incident has been resolved
resolution_date	date	Date the incident was resolved (nullable)
reported_by	string	Name of the staff member who reported the incident
follow_up_required	boolean	Whether follow-up action is required

social_media_posts

Records of the organization's social media activity across platforms, modeled after the data available through platform APIs (Twitter/X, Facebook Graph, Instagram Insights, TikTok, LinkedIn, YouTube Data,

WhatsApp Channels). Each row represents a single post with its content, metadata, and engagement metrics. Used to analyze social media effectiveness and its relationship to donor behavior.

Field	Type	Description
post_id	integer	Primary key
platform	string	One of: Facebook, Instagram, Twitter, TikTok, LinkedIn, YouTube, WhatsApp
platform_post_id	string	Simulated platform-native post ID (e.g., fb_1234567890123456)
post_url	string	Permalink URL to the post
created_at	datetime	Full date and time the post was published (includes hour and minute)
day_of_week	string	Day of the week (e.g., Monday, Tuesday)
post_hour	integer	Hour of the day the post was published (0–23)
post_type	string	One of: ImpactStory, Campaign, EventPromotion, ThankYou, EducationalContent, FundraisingAppeal
media_type	string	One of: Photo, Video, Carousel, Text, Reel
caption	string	Full text caption of the post
hashtags	string	Comma-separated list of hashtags used (e.g., #EndTrafficking, #HopeForGirls)
num_hashtags	integer	Count of hashtags used
mentions_count	integer	Number of @mentions in the post
has_call_to_action	boolean	Whether the post includes a call to action
call_to_action_type	string	One of: DonateNow, LearnMore, ShareStory, SignUp (nullable)
content_topic	string	One of: Education, Health, Reintegration, DonorImpact, SafehouseLife, EventRecap, CampaignLaunch, Gratitude, AwarenessRaising
sentiment_tone	string	One of: Hopeful, Urgent, Celebratory, Informative, Grateful, Emotional
caption_length	integer	Character count of the caption
features_resident_story	boolean	Whether the post features a specific resident's anonymized story
campaign_name	string	Associated campaign name, if any (nullable). Uses the same campaign names as donations.campaign_name
is_boosted	boolean	Whether paid promotion was used for this post
boost_budget_php	decimal	Amount spent on paid promotion in Philippine pesos (PHP) (nullable; only populated if is_boosted is true)
impressions	integer	Total number of times the post was displayed
reach	integer	Number of unique accounts that saw the post
likes	integer	Number of likes or reactions
comments	integer	Number of comments
shares	integer	Number of shares or retweets
saves	integer	Number of saves or bookmarks
click_throughs	integer	Number of clicks to the organization's website

video_views	integer	Number of video views (nullable; only populated for Video and Reel media types)
engagement_rate	decimal	Engagement rate: (likes + comments + shares) / reach
profile_visits	integer	Number of profile visits attributed to this post
donation_referrals	integer	Number of donations attributed to this post
estimated_donation_value_php	decimal	Estimated total Philippine pesos (PHP) value of donations referred by this post
follower_count_at_post	integer	Organization's follower count on this platform at the time of posting
watch_time_seconds	integer	Total watch time in seconds across all viewers (nullable; only populated for YouTube posts)
avg_view_duration_seconds	integer	Average number of seconds each viewer watched (nullable; only populated for YouTube posts)
subscriber_count_at_post	integer	YouTube subscriber count at the time of posting (nullable; only populated for YouTube posts)
forwards	integer	Number of message forwards (nullable; only populated for WhatsApp posts). WhatsApp forwards represent personal referrals with high donation conversion rates

safehouse_monthly_metrics

Pre-aggregated monthly metrics for each safehouse.

Field	Type	Description
metric_id	integer	Primary key
safehouse_id	integer	FK → safehouses.safehouse_id
month_start	date	First day of the month
month_end	date	Last day of the month
active_residents	integer	Number of residents assigned during the month
avg_education_progress	decimal	Mean education progress (0–100)
avg_health_score	decimal	Mean general health score (1.0–5.0)
process_recording_count	integer	Total process recording entries for the month
home_visitation_count	integer	Total home visitations conducted
incident_count	integer	Total safety incidents recorded (from incident_reports)
notes	string	Free-text notes

public_impact_snapshots

Monthly anonymized aggregate impact reports intended for public-facing dashboards and donor communications.

Field	Type	Description
`snapshot_id`	integer	Primary key
`snapshot_date`	date	Month the snapshot represents
`headline`	string	Headline text for the snapshot
`summary_text`	string	Short paragraph summary
`metric_payload_json`	string	JSON string of aggregated metrics
`is_published`	boolean	Whether the snapshot has been published
`published_at`	date	Publication date